

Lem diferencia productos leq suma diferencias

Lema 1. Sean $\{a_k\}_{k=1}^n, \{b_k\}_{k=1}^n \subset \overline{\mathbb{D}}$. Entonces, $\left| \prod_{k=1}^n a_k - \prod_{k=1}^n b_k \right| \leq \sum_{k=1}^n |a_k - b_k|$.

Demostración: Lo probaremos por inducción en n . Para $n = 1$, el resultado es trivial. Supongamos que el resultado se cumple para n . Consideramos $\{a_k\}_{k=1}^{n+1}, \{b_k\}_{k=1}^{n+1} \subset \overline{\mathbb{D}}$. Entonces,

$$\begin{aligned}
 \left| \prod_{k=1}^{n+1} a_k - \prod_{k=1}^{n+1} b_k \right| &= \left| \left(\prod_{k=1}^{n+1} a_k - a_{n+1} \prod_{k=1}^n b_k + a_{n+1} \prod_{k=1}^n b_k - \prod_{k=1}^{n+1} b_k \right) \right| \\
 &= \left| a_{n+1} \left(\prod_{k=1}^n a_k - \prod_{k=1}^n b_k \right) + \left(\prod_{k=1}^n b_k \right) (a_{n+1} - b_{n+1}) \right| \\
 &\leq |a_{n+1}| \left| \prod_{k=1}^n a_k - \prod_{k=1}^n b_k \right| + \prod_{k=1}^n |b_k| |a_{n+1} - b_{n+1}| \\
 &\leq \left| \prod_{k=1}^n a_k - \prod_{k=1}^n b_k \right| + |a_{n+1} - b_{n+1}| \leq \sum_{k=1}^n |a_k - b_k| + |a_{n+1} - b_{n+1}|. \quad \blacksquare
 \end{aligned}$$

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